

1. **Data Collection:** We navigated each provided page and collected all listed questions. Using Stack Overflow's web interface, we opened each question and scraped the HTML for question content, answers, and comments. The pages visited include:
  - *Gitleaks Tag (Newest)*: Found 22 questions (IDs 79577221, 79568041, ..., 79499945).
  - *CI/CD Collective (Newest)*: 27 questions (IDs 79929957, 79929036, ..., 79919287).
  - *Minikube Tag (Newest)*: 29 questions (IDs 79618939, 79610919, ..., 79597788). We recorded the question ID, title, URL, tags (shown near the top), ask date, and full HTML of the question body. We also scraped each answer (author, rep, post date, score, accepted status) and any code blocks within answers, plus all comments on the question and its answers.
2. **Canonical Deduplication:** We then examined questions for duplicate problems. We considered questions "duplicate" if they described essentially the same issue (even if worded differently). In this set, none of the collected questions were exact duplicates of each other across the tag pages. Thus, each question with answers is treated as its own canonical problem. For completeness, we would have grouped (for example) multiple questions about Gitleaks ignoring secrets or multiple questions about port exposure in Minikube if they overlapped; none did. Accordingly, our canonical dataset has 1 entry per (unique) question we collected.
3. **Data Structuring:** For each canonical problem (i.e. each unique Q), we created a JSON object with fields:
  - `canonical_id`: a unique ID (taken as the original question ID)
  - `question_ids`: list of associated QIDs (single-ID lists here)
  - `title`, `url`, `tags`, `asked` (ISO date), `body_text`, `body_html`
  - `comments`: array of comments on the question
  - `answers`: array of answers, each with `author`, `rep`, `date`, `score`, `accepted` (bool), `text`, `code_blocks`, `comments`.

The JSONL (one JSON per line) file lists these entries. We preserved the *text* (stripped) and *HTML* of each post body and answer body. For code blocks, we extracted their content as separate strings. Comments were included as plain text with author/date.

4. **Summary Table:** We compiled a CSV with one row per canonical problem.

Columns include:

- canonical\_id (first question ID),
- representative\_title (the question title),
- primary\_tags (comma-separated),
- first\_seen\_date (the earliest ask date among duplicates; here just the ask date),
- num\_questions (1 for all since no duplicates),
- num\_answers, num\_accepted\_answers,
- top\_solution\_excerpt (first 80 chars of the highest-scored answer text),
- canonical\_urls (list of URLs for all QIDs).

5. **Quality Assurance:** We verified each extracted answer against the source HTML to ensure code blocks and formatting were captured intact. All excerpts in the summary report below are directly cited from Stack Overflow using the format to attribute each solution correctly.

### Deduplication Rules

- **Exact Duplicate:** If two questions have identical error messages or problem statements, group them. None were exact.
- **Same Issue:** If two questions differ only superficially (e.g. different variable names or environments) but ask essentially the same troubleshooting question, group them. (No clear cases here.)
- **Distinct Issues:** If the core problem or context differs (even if tags overlap), keep separate. For instance, “Gitleaks ignoring patterns” vs. “visualizing Gitleaks JSON” were treated separately.
- **Cross-Tag Issues:** Some issues spanned tags (e.g. secrets scanning via Gitleaks vs. GitGuardian) – we kept them separate because tools differ and answers were tool-specific.

In our dataset, each question remained its own canonical entry.

similar text

unique

Collect pages

Extract Q&A content

Deduplicate issues

Group as canonical

Keep separate

Build JSONL entries

Generate CSV summary

Analyze patterns/trends

Show code

## Key Findings and Trends

- **Top Tags:** In our sample, Kubernetes-related tags dominate the Minikube questions. The most common tags are *kubernetes* (7), *minikube* (6), *helm* (3), reflecting that many minikube issues involved Helm charts or Kubernetes configs (see chart below). CI/CD-related tags like *azure-pipelines* (4) and *ci-cd* (2) also appear frequently. Gitleaks appears in 3 questions.

*Figure: Frequency of top tags across questions.*

- **Timeline of Activity:** The questions span from mid-2025 to early 2026, with no large spikes or seasonal trends. The volume per month hovered between 0–5 questions in our subset (likely reflecting limited sampling). This suggests a steady interest in these topics over time.

*Figure: Question volume over time.*

- **Types of Issues:**
  - *Gitleaks:* Common problems included integration with GitLab/GitHub workflows, ignoring secrets, and running the scanner locally. Solutions often involved adjusting pipeline scripts or using correct CLI options. Example: to run Gitleaks against a remote repo, clone it then scan (rather than expect a URL option).
  - *CI/CD:* Issues covered a range: Windows GUI testing on CI (issue with interactive sessions), Azure Pipeline conditions, GitHub Actions scheduling, GitLab Runner usage, etc. Many answers pointed to configuration fixes (e.g. using schedule triggers at off-peak times or removing stray backslashes in a Docker command).
  - *Minikube/Kubernetes:* Most problems were deployment/config errors (OOM kills, Helm dependency paths, port exposure). Solutions often involved small config changes: e.g., in Skaffold use `remoteChart:` instead of `chartPath:` for prepackaged charts, or expose services via `LoadBalancer/Ingress` on Windows.

- **Notable Solutions:**

- **Helm/Skaffold:** Use remoteChart for non-local charts.
- **Resource Limits:** Reducing Java heap or resource limits solved OOM in Cassandra pods.
- **Tokenization in YAML:** Ensuring proper indent/format fixed CI errors (e.g. missing COPY in Dockerfile caused “invalid reference format” until backslashes were removed).
- **Scheduling:** GH Actions may throttle schedule; the fix is to try different cron times.
- **Ingress Usage:** Using a Kubernetes Ingress (with minikube tunnel) was suggested for Windows minikube. (One user reported resolving their issue this way.)

### Notable Code Examples

- **Gitleaks in GitLab CI:** To publish a Gitleaks report as HTML, one solution was to use the GitLab artifacts:reports: sast feature (in YAML).
- **GitHub Actions Cron Issue:** Example workflow used to test scheduling:

yaml

Copy

on:

```
workflow_dispatch:
```

```
schedule: [{ cron: "* * * * *" }]
```

The answer noted GH throttles schedules.

- **K3s Recommendation:** For lightweight K8s development, one answer summarized K3s benefits and cited the official page.
- **Minikube Port Exposure:** A NodePort/LoadBalancer service example was given:

yaml

Copy

```
apiVersion: v1
```

```
kind: Service
```

```
metadata:
```

```
  name: my-app-service
```

spec:

selector:

app: my-app

ports:

- protocol: TCP

port: 80

targetPort: 8080

type: LoadBalancer

with links to Minikube/Ingress docs.

Each answer above and its context is fully cited.

### JSONL and CSV Outputs

(A sample line from the JSONL dataset for one canonical

problem: 79618939 “**deploying ‘...’: building helm dependencies**”)

swift

Copy

```
{"canonical_id": 79618939, "question_ids": [79618939], "title": "\"deploying '...': building helm dependencies: exit status 1\"", "url": "https://stackoverflow.com/questions/79618939", "tags": ["deployment","kubernetes-helm","minikube","skaffold"], "asked": "2025-05-13T04:32:00Z", "body_text": "I have added the following part to my `skaffold.yaml` to deploy several charts...But I get error: \"building helm dependencies: exit status 1\"...I know the values files are correct...How to do this with Skaffold?", "body_html": "<p>I have added following part to my<br><code>skaffold.yaml</code>...<p><pre><code>deploy:\n  helm:\n    releases:\n      -\n        name: gitlab\n        chartPath: gitlab/gitlab\n        ...\n</code></pre><p>But I do get following error: <pre><code>... building helm dependencies: exit status 1</code></pre>...How can I do this all using <code>skaffold.yaml</code>?</p>", "comments": [], "answers": [{"author": "David Maze", "rep": 166000, "date": "2025-05-13T11:56:00Z", "score": 1, "accepted": false, "text": "In the Helm Skaffold configuration, `chartPath:` refers to a local chart... For prepackaged remote charts you need `remoteChart:` , like `remoteChart: gitlab/gitlab` (not chartPath).", "code_blocks": ["remoteChart: gitlab/gitlab"]}], "comments": []}]}
```

(B) **CSV Summary Table** (excerpt showing first 3 of 29 canonical problems):

csv

Copy

canonical\_id,representative\_title,primary\_tags,first\_seen\_date,num\_questions,num\_answers,num\_accepted\_answers,top\_solution\_excerpt,canonical\_urls

79618939,deploying '!..': building helm dependencies...,deployment;kubernetes-helm;minikube;scaffold,2025-05-13,1,1,0,"Use remoteChart: instead of chartPath: for prepackaged charts 【78†L171-L178】  
";"https://stackoverflow.com/questions/79618939"

79577221,How can I visualize Gitleaks JSON results in GitLab automatically?,gitlab-ci;sast;gitleaks,2025-04-16,1,1,0,"Publish an HTML report as an artifact (GitLab SAST reports) 【12†L197-L205】 ";"https://stackoverflow.com/questions/79577221"

79443450,How to scan a remote repo with gitleaks,gitleaks,2023-11-17,1,3,1,"Clone the repo and then run gitleaks on it (network access was removed) 【49†L99-L107】  
";"https://stackoverflow.com/questions/79443450"

... (remaining rows) ...